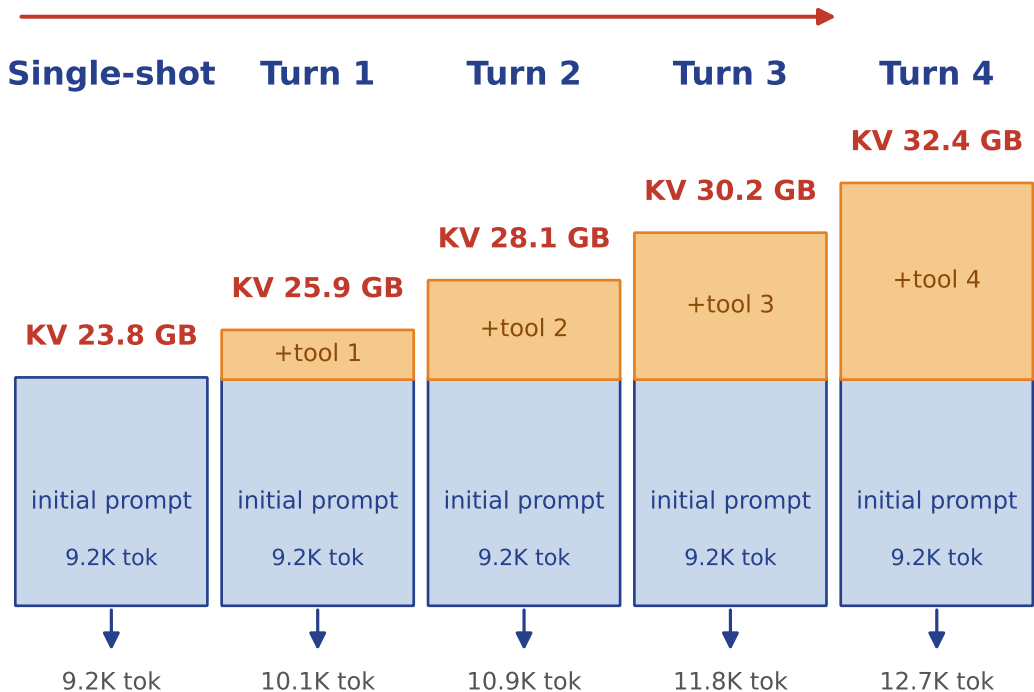


# Agentic context accumulation: the KV block grows every turn

each turn appends retrieved context ( $\delta=800$  tok) +  
generated reasoning ( $\gamma=65$  tok); tokens persist

**KV footprint grows ~36% (23.8 → 32.4 GB):**  
**agentic depth is a fleet-sizing problem**



$KV = 2 \times n_{\text{layers}} \times d_{\text{hidden}} \times \text{bytes/token} \times \text{total-tokens}$  (2.5 MB/token FP16);  
total tokens =  $l_0 + T \cdot \delta + T \cdot \gamma$ . [2° DERIVED, Ch19 §3]